SPACE MAGNUM EQUIPMENTS PVT. LTD.

Storage ideas resulting into:

Capacity² Efficiency² Safety²



STOLIFT



STOMAT



STOPICK AS/RS



Storage ideas resulting into **Capacity² Efficiency² Safety²**



SPACE MAGNUM EQUIPMENTS PVT. LTD.

A few words about us

In 1992, Space Magnum started as a small fabrication and assembly shop undertaking job work for conventional Mobile Racks.

While doing this, from a close interaction with several users of storage systems it was revealed that there are several manufacturers of a variety of storage systems right from slotted angle racks to automated vertical carousels. But, almost none were considering the stores in totality. In several cases, in their attempt to 'sell' whatever they produced or had, the customers were paying a hefty price for 'wrong products'.

The situation definitely called for a wider perspective and integrated approach. The thought proved an inspiration for Space Magnum.

With a higher thrust on product development and after having developed innovative solutions for over a decade, today Space Magnum is equipped with a wide range of storage systems for a variety of storage needs ranging from 0.5 gm washer to 5 Ton injection molding die and from 1 cm^3 IC chip to 1.5 m^3 corrugated boxes.

Over the years, the company has not only laid emphasis on the development of technologically advanced storage systems, but also mastered in several other allied areas such as system study, project management, structural engineering, electronics engineering and software development for storage systems. With its sincere approach and committed efforts, today Space Magnum has acquired a reputation among its customers of being a 'Storage System Solution Provider.'

The willingness to develop customised solution, designing and manufacturing a flawless product and honouring time commitments culminate into a wide customer base all over the country and in a variety of industries.

For Space Magnum, this is just the beginning of revolutionising the concept of stores. Or should we call it TSM?

Together, we will make it happen.

Sasewadi, Pune
Plant - II



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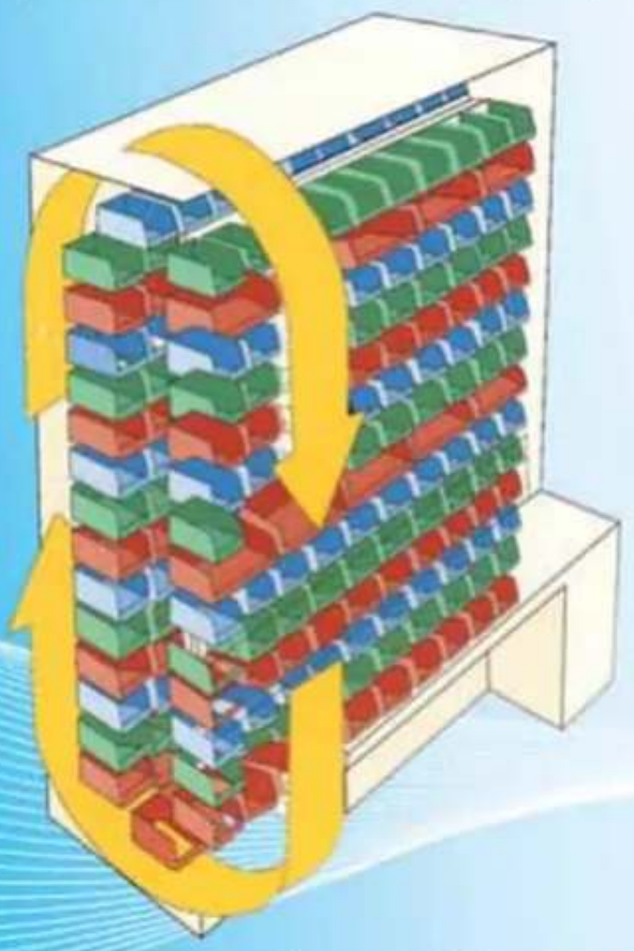
Stomat

Vertical Carousel Storage System

Automated Vertical Carousel

This is an advanced storage and retrieval system wherein the required material is retrieved by just pressing a button on the key board.

Operates on Paternoster elevator principle



The concept

In its simplest form, a number of shelves or material storing trays are mounted on vertical chain loops and provided with a PLC controller. The desired rack can be retrieved by entering the shelf number. While retrieving, it takes the shortest possible route. The entire system is encased in a sheet metal cladding.

Where to use ?

Ideal system for small parts storage where the number of items to be stored is very high.

Benefits

Quick retrieval

Saves cost, time, energy three times than any conventional system

Space saving

The floor space is saved by 40% and the available height is used to its maximum

Manpower saving

Precious manpower is saved due to highest automation

Low noise

Electronically controlled and operated by geared chains, so less noise

Easy to operate

Material retrieval at the touch of a button. Can be operated manually in case of power failure

Operator safety

Maximum operator-friendly system. Electronic sensors stop the system in the possibility of malfunctioning

Custom-built system

Designed and manufactured taking into account various requirements of different customers

Storage ideas resulting into: Capacity² Efficiency² Safety²



SPACE MAGNUM EQUIPMENTS PVT. LTD.

Stomat

Vertical Carousel Storage System

Constructional features

- Vertical carousel in which a number of carriers are closely mounted on the twin endless chain moved by a geared brake motor
- Operates on Paternoster elevator principle
- The mechanised shelves rotate in the vertical plain in either direction
- The material reaches the operation zone by the nearest path
- Torsion-free, self-supporting, powder-coated construction
- Work table at 1 meter from the ground
- Provided with PLC controllers

Sophisticated controller

- Provided with an advanced, microprocessor controller with numeric display
- Desired tray, bin or component can be retrieved by its part number
- In case of power failure, present position of the carrier is memorised by PLC
- Can be interfaced with a PC or control computer system through RS 232 C serial port

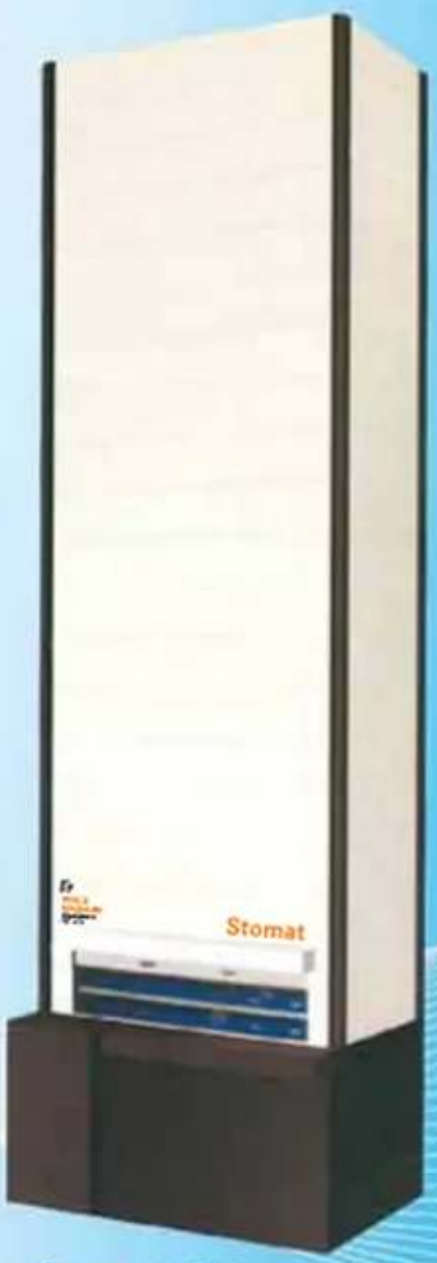
Safety features

- Safety bars are provided at the top and bottom side of the operating window. In case of accidental pressing of these bars, the system will automatically stop
- The system can be operated manually in case of power failure

Typical applications

Ideal for storing

- Tools
- Hardware
- Components
- Office files
- Documents
- Machined components
- Electronic parts
- CDs and cassettes
- Medicines
- Small, costly finished goods etc.



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STOMAT (VERTICAL CAROUSEL)









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STOMAT (VERTICAL CAROUSEL)

Model	Number of Shelves	Machine Height (mm)	Shelf Height (mm)	Loading capacity per shelf (Kg)	Total load Capacity (Kg)
SM - 650					
65-135-24	24	2235	135	65	1560
65-135-36	36	3150	135	65	2340
65-135-49	49	4140	135	65	3185
65-135-57	57	4750	135	65	3705
SM - 1400					
140-185-24	24	3028	185	140	3360
100-185-34	34	4044	185	100	3400
100-185-44	44	5060	185	100	4400
140-260-18	18	3163	260	140	2520
140-260-24	24	4002	260	140	3360
100-260-32	32	5110	260	100	3200
140-325-12	12	2850	325	140	1680
140-325-18	18	3916	325	140	2520
140-325-24	24	4971	325	140	3360
SM - 2400					
240-220-23	23	3950	220	240	5520
220-220-27	27	4475	220	220	5940
170-220-32	32	5140	220	170	5440
150-220-36	36	5680	220	150	5400
250-258-20	20	3940	258	250	5000
240-258-24	24	4530	258	240	5760
200-258-28	28	5150	258	200	5600
170-258-32	32	5750	258	170	5440
250-373-17	17	4550	373	250	4250
250-373-20	20	5140	373	250	5000
240-373-23	23	5800	373	240	5520

Horizontal Carousel

Triple, horizontal carousel for storage of electronic components.



Multi-loop Horizontal Carousel

for different sizes of metal profiles.

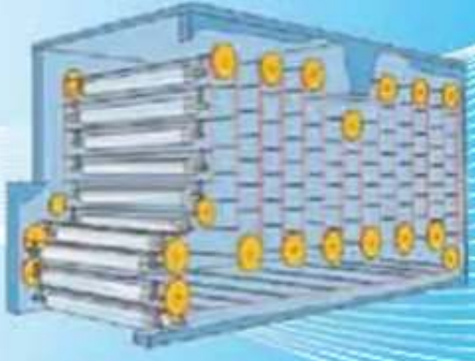


Technical Data

PARTICULARS	SM - 650 MODEL	SM - 1400 MODEL	SM - 2400 MODEL
FLOOR AREA			
Length	2500 mm	2700 mm	2750 mm
Depth	900 mm	1400 mm	1800 mm
WORK TABLE			
Depth	300 mm	300 mm	300 mm
Height	1000 mm	1000 mm	1000 mm
(from ground)			
USEFUL SHELF DIMENSIONS			
Length	1970 mm	2160 mm	2160 mm
Depth	230 mm	420 mm	520 mm
Necessary Room Height	**Machine height plus 100mm**		

Multi-loop Vertical Carousel

for storage of print cylinders.



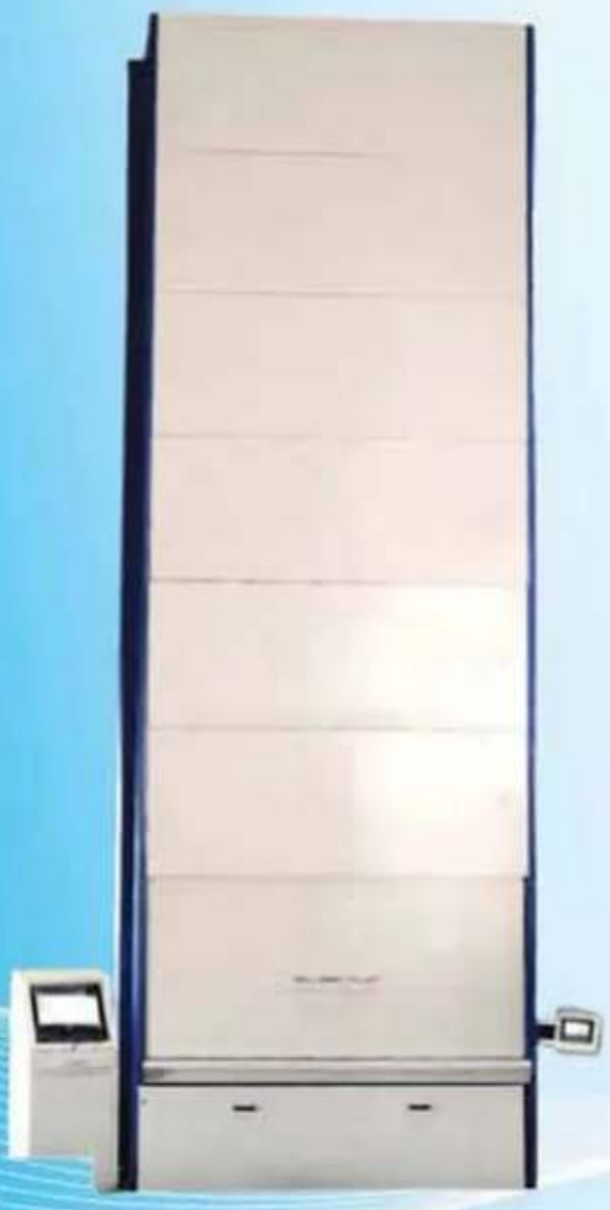
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Stolift

Modular Automated Storage & Retrieval System (Shuttle)

This type of storage system is a recent development combining advantages of several other systems. It is compact, sturdy and ensures fast retrieval.



Simple yet powerful Vertical lift technology with the Stolift from Space Magnum

Storage ideas resulting into: Capacity² Efficiency² Safety²

The concept

It is an advanced and powerful storage system provided with sophisticated PLC controller and yet very easy to operate. PLC controlled platform brings special material carriers from the control station to a particular storage location & vice-versa.

At the operating window, material can be placed on to the retriever tray. When its location / part number is entered, it will be automatically placed at the pre-determined location in the storage rack. For material retrieval, entering the part number activates the retriever, which reaches the designated rack and retrieves it to the operating window. Very simple and very fast.

Where to use ?

The system is ideal for mid-sized and mid-weight components / items / parts which need safe storage, fast retrieval and precision inventory control. And also, the height utilisation is possible and required.

Advantage over vertical carousel system

- No load imbalance
- Pitch of trays can be altered and is flexible
- Multiple pitches can be provided
- Fixed pitch or optimized pitch (Machine automatically allocates space, considering the component height)
- Flexo at 75 mm to 900 mm pitch
- Machine height can be increased at a later date
- Power consumption is very less compared to vertical carousel
- Capacity can be increased at a later date
- No vibration or jerks during the operation
- Machine capacity is designed for 20 Tonne and 40 Tonne
- Material can be taken out from the machine along with the trays
- Further automation is possible for transferring of materials through material handling systems
- Bar code scanner can be connected externally to the machine
- Material can be accessed from the top (easy accessibility)
- Auto Door Provided
- Work table weight carrying capacity facility available



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Stolift

Modular Automated Storage & Retrieval System (Shuttle)

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Benefits

Quick retrieval

Since only the retriever moves to locate and retrieve tray / material, the speed can be higher, ensuring fast retrieval.

Food space saving

Floor space can be saved to a great extent (40%) since the entire available height is being used and even the operating windows can be provided at 2 or 3 different levels.

High safety of material

Material is stored in completely enclosed racks, ensuring safety from theft, dust, mishandling or damage.

Manpower saving

Due to complete automation and speedy retrieval, manpower is saved to a considerable extent.

Inventory control

Material is retrieved only after entering the part number & quantity and hence, precise control on the inventory can be achieved.



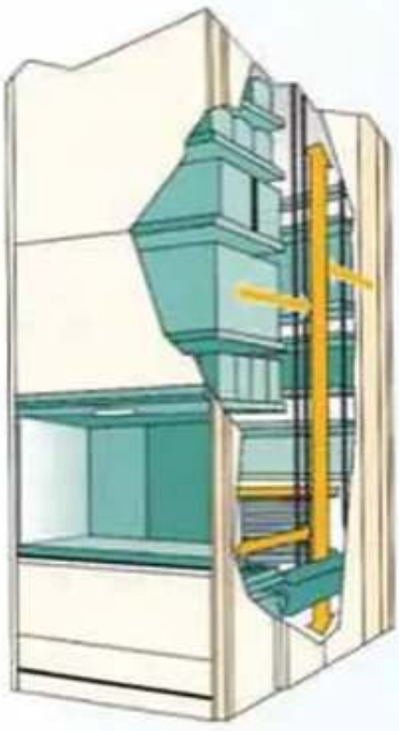
Storage ideas resulting into: Capacity² Efficiency² Safety²

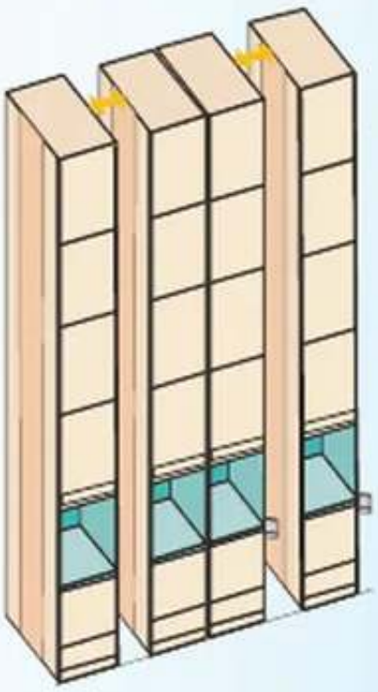


SPACE MAGNUM EQUIPMENTS PVT. LTD.

Stolift

Modular Automated Storage & Retrieval System (Shuttle)







Trays with plastic boxes



Trays with modular side-walls, partitions and dividers



Trays for Euro-pallets



Trays with modular side-walls and plain partitions



Trays with a tool holder system

Tray Size	Tray Capacity	Machine Height
620 x 1220	250Kg/500Kg.	
1550	250Kg/500Kg.	
1850	250Kg/500Kg.	
2150	250Kg/500Kg.	
2440	250Kg/500Kg.	
2750	250Kg/500Kg.	
2950	250Kg/500Kg.	
825 x 1220	250Kg/500Kg.	
1550	250Kg/500Kg.	
1850	250Kg/500Kg.	
2150	250Kg/500Kg.	
2440	250Kg/500Kg.	
2750	250Kg/500Kg.	
2950	250Kg/500Kg.	

Machine height depends on your ceiling ht & tray Qty depends on the material height, and Flexo at 75 mm min 150 mm.

Storage ideas resulting into: Capacity² Efficiency² Safety¹



Storder

Modular Multi-axis Storage & Retrieval System

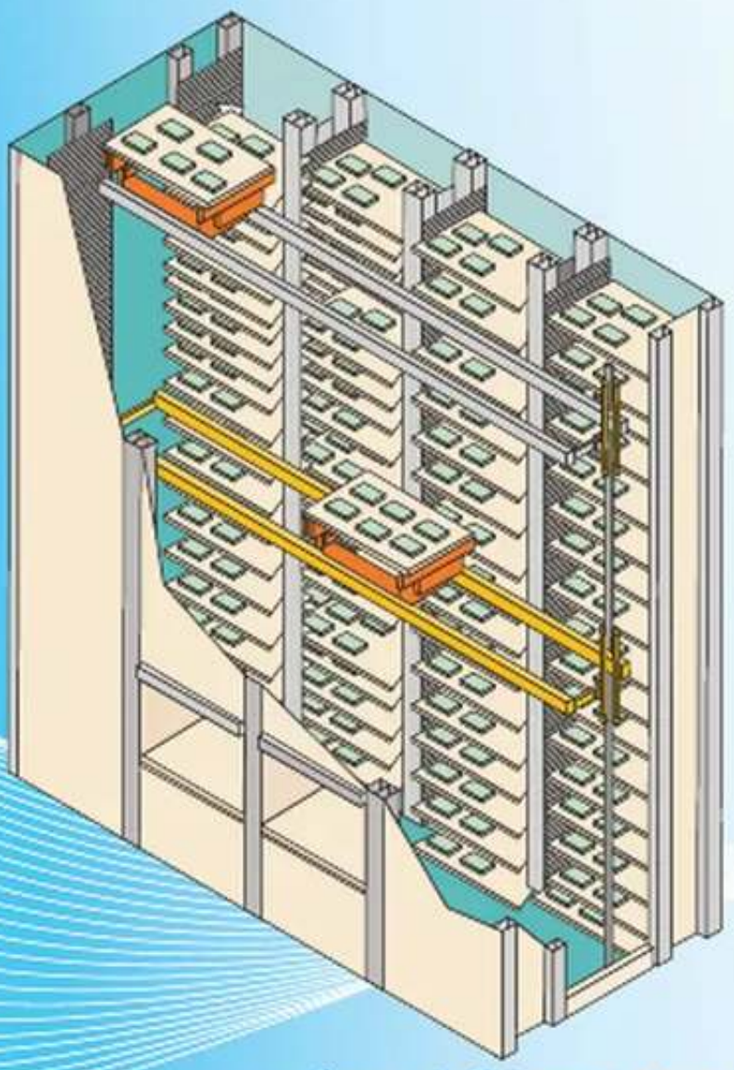
Storder is the latest addition to the product range of Space Magnum. This new-generation storage system utilizes a triple-axis (X, Y, Z) movement.

Storder assures optimized efficiency & storage capacity as it blends the advantages of both Stolift & Stopick in a limited space. It is most effective where a large number of SKUs (Stock Keeping Units) cannot be accommodated in a single Stolift machine without hampering the accessability of the SKUs.

Storder can be expanded only vertically by increasing the number of windows.

The number of picks per hour can be increased or matched with the number of

picks of individual Stolift machine.



Benefits:

- Storder estimates the need for access openings on each machine.
- It reduces cost of maintenance and number of operations.
- It can achieve a high level precision and automation of storage & retrieval
- needs of manufacturing industry, stocklist, distributions, warehouses.

Special Features

- Retrieving mechanism of unique design which is light weight & yet robust
- High-performance chain with less noise during operation
- Torsion-free structure & sturdy covers
- Excellent positioning accuracy

Storage ideas resulting into: Capacity² Efficiency² Safety²



SPACE MAGNUM EQUIPMENTS PVT. LTD.

Stopick AS/RS

Automated Storage & Retrieval System

This is the most modern unit load, storage and retrieval system, combining advantages of several systems. It is simple, efficient, modular and easily expandable system. It not only utilizes full height, but also uses the depth to any extent.





Storage ideas resulting into: Capacity² Efficiency² Safety²



Stopick AS/RS

Automated Storage & Retrieval System

Wide ranging applications

This system is highly versatile and the application areas are countless. It can be practically used for storing every type of material right from small electronic parts, hardware, small tools to refrigerators, drums in chemical / process industries and even storing & retrieving passenger cars in parking lots.



Storage ideas resulting into: Capacity² Efficiency² Safety²

Constructional Features

- Positive hoisting by chain
- High speed application. 24 to 40 dual cycles per hour
- Simultaneous X & Y axis motion
- Current through shrouded bus bars/ leading trolley, trailing trolley
- On-board SRM control communicates with off-board computer
- On-board diagnostics identify SRM faults for rapid repair and recovery
- Double arm lifting carriage ensures stability/rigidity of system
- Lifting carriage is guided on 8 rollers, ensures smooth and quiet operation
- **Load extraction - multiple choices e.g.**
  - **Single arm telescopic fork**
  - **Double arm telescopic fork**
  - **Retriever I (*TM)**
- Core sensing ensures error-free operation
- Ergonomically designed work station
- **Aisle changer hardware for multiple aisle racks**
- The drive and control panel are mounted at floor level for easy access
- High rate of acceleration
- Machine height up to 16 meters (Lattice structure mast)
- Hoisting and long travel are achieved through AC variable frequency drive, ensures precision and smooth operation
- Quadruple chain hoisting ensures complete safety
- Linear guide ways for extractor **Retriever I**
- Optical load centering device on board
- Race supported structures ensure lower building cost
- Optional man on-board available
- Can be easily interfaced with a PC or central computer system

Safety features

- Full safety protocols through on-board sensors as per FEM 9.754
- Racking structure, stringent design, standard as per 9.831 requirement SRM as per FEM 9.311



Stopick AS/RS

Automated Storage & Retrieval System



The concept

The total system consists of SRM (Storage Retrieval Machine) wherein the load carrier mounted on the mast and guided by rails moves in horizontal and vertical plain between the aisle of two storage racks. The operator window is provided at the front end of the aisle. The system is provided with a powerful PLC. The light weight and sturdy construction does not impose any limitation on the height and depth of the storage but also the speed of the load carrier can be selected depending on the requirement. Simultaneous movement of the load carrier in vertical and horizontal plane ensures high speed retrieval. This completely customised system ensures complete automation, high density storage, quick retrieval and safety.

Due to several such advantages, AS/RS are popular in developed countries and gaining acceptance over other systems even in India.

Where to use ?

Being highly customised, they can be practically used in any and every stores. However, typical applications are in warehouses, finished goods and raw material stores in a large manufacturing unit such as pharmaceutical, automobile, engineering, heavy electricals etc.



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Stopick AS/RS

Automated Storage & Retrieval System

Benefits

Cost-effective system

Single retrieving mechanism for a large storage area saves on the cost of the system.

Integral storage

Single, integral storage area due to optimal utilisation of height and depth ensures high density storage and allows easier control.

Floor space saving

Complete height utilisation, extreme narrow aisle for column movement ensures considerable floor space saving.

One man operation

One person can operate the system very easily since storage and retrieval is completely automated.

Fast retrieval

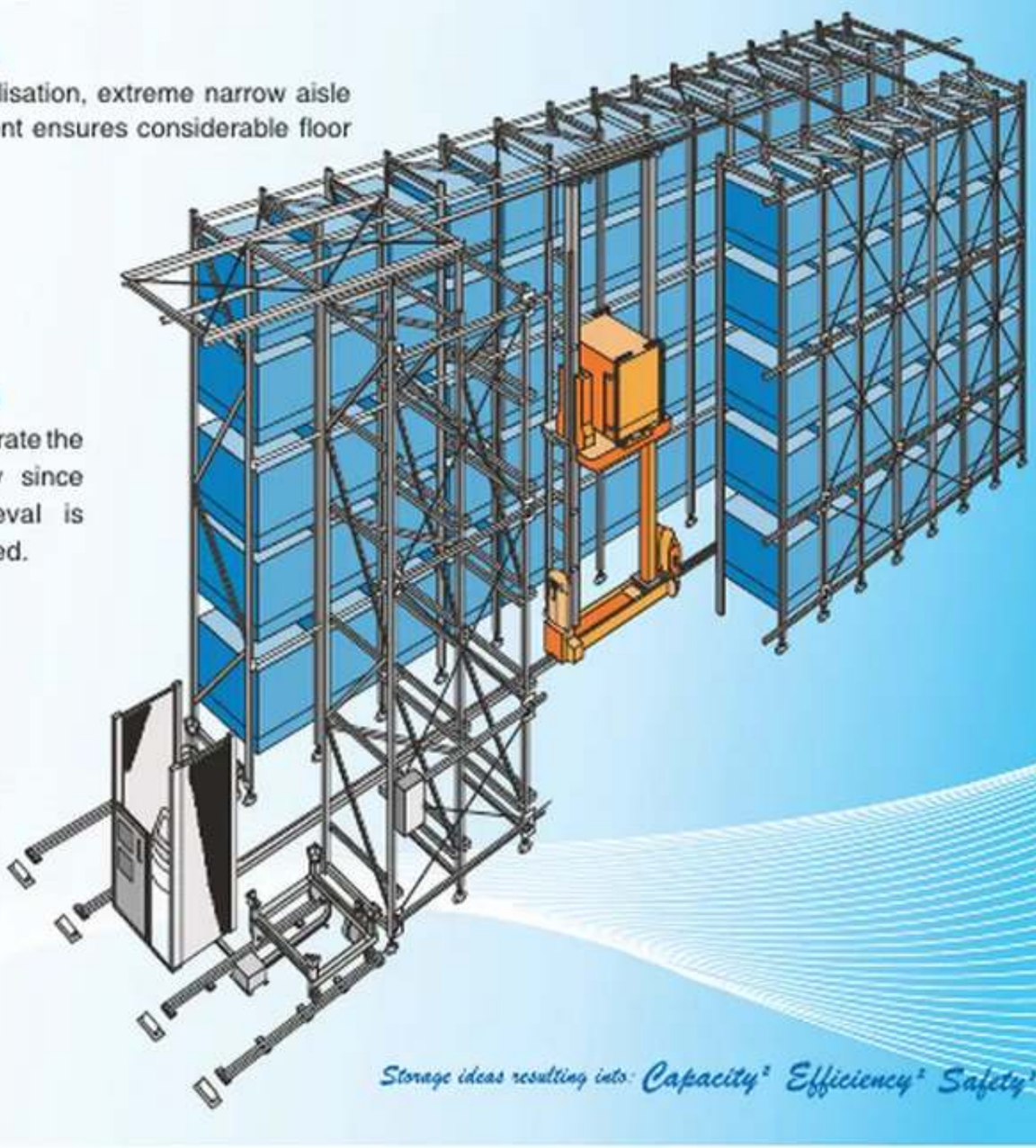
High speed and simultaneous movement in both the plains ensure fast retrieval of material.

High safety of stored material

Completely enclosed system with locking of operating window.

Easily expandable

Modular system allows easy expansion of racks and main carriage being the same, the additional cost is comparatively lesser.



Storage ideas resulting into: Capacity² Efficiency² Safety²



Stomat Compactors

Mobile Rack Storage System





The system that knows no limits

Stomat mobile shelving provides extensive storage space within a limited area. And if the need for more storage becomes evident at a later date,

Stomat has the solution.

More space, and no need to expand. That cuts costs. The Stomat compactor system requires far less room than conventional shelving and with new premises you can even allow for extra spaces in the planning phase.

Stomat compactor, the right system for every application.

Mobile shelving - Benefits

Instead of many space-consuming, individual aisles between fixed units, with the Stomat mobile system, a single aisle provides access to all shelves. Fixed shelving is only needed for simultaneous access to all items in the store. In all other cases, aisles between each shelving unit are simply uneconomical. So just push your shelves together. It's as easy as that.

Whether in offices, workshops or households, Stomat is the ideal solution. Practical examples show that you can store while saving space and still maintain rapid access. Files, suspended folders, registers, card indexs, clothes, fittings, electrical and electronic accessories, shoes, paints, medicaments, bins and a lot more. Stomat compactor demands little space and offers a storage capacity that you would never have thought.

Storage ideas resulting into: Capacity² Efficiency² Safety²



SPACE MAGNUM EQUIPMENTS PVT. LTD.

Stomat Compactors

Mobile Rack Storage System

Optimal solution for all types of storage

Aisle security

For your special requirements or tailor-made products, call on our trained advisors for a complete site survey, design, supply and installation service.

Drives

We can provide suitable drives for every weight class and application to suit the overall weight of your storage shelves and their frequency of use.

Manual operation

A block of several shelving base with a maximum overall weight of 1200 kg can be moved using the manual system

Crank operation

The crank drive with a 1:6 reduction ratio permits easy movement of shelving base or blocks of bases with a maximum weight of 10 Tonne

Electric drive

This drive is mainly used in systems with very heavy loads. Single motor drive is via a friction wheel running on the rail. The control system is designed to meet specific customer requirements.

Central locking system

The central locking system protects the entire system against unauthorised access.









Storage ideas resulting into Capacity² Efficiency² Safety¹



Stomat Compactors

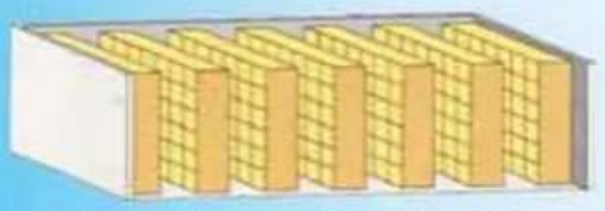
Mobile Rack
Storage System

Stomat Compactors
Mobile Rack Storage System

Ht. 2040	300 SFF	375 SFF	450 SFF	600 DFM	750 DFM	900 DFM
900 width 1800(W) 2700(W) 3600(W)	5 levels of storage Suitable for box files/latra filing					
Ht.2440	**300 SFF**	**375 SFF**	**450 SFF**	**600 DFM**	**750 DFM**	**900 DFM**
900 width 1800(W) 2700(W) 3600(W)	6 levels of storage Suitable for box files/latra filing					
Ht.2840	**300 SFF**	**375 SFF**	**450 SFF**	**600 DFM**	**750 DFM**	**900 DFM**
900 width 1800(W) 2700(W) 3600(W)	7 levels of storage Suitable for box files/latra filing					

Fixed units with 30 bays,each providing six meters of shelf space = 180 linear meter With mobile shelving , the same space can house 54 shelving units with the same shelf space = 324 linear meters of shelf space. The gain in space is thus 144 linear meters - an improvement of 80%!

Conventional Racking System



Extra working area made available : **24%**

Mobile Shelving



Increase in storage capacity around **83%**



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SPACE MAGNUM EQUIPMENTS PVT. LTD.

Stolong

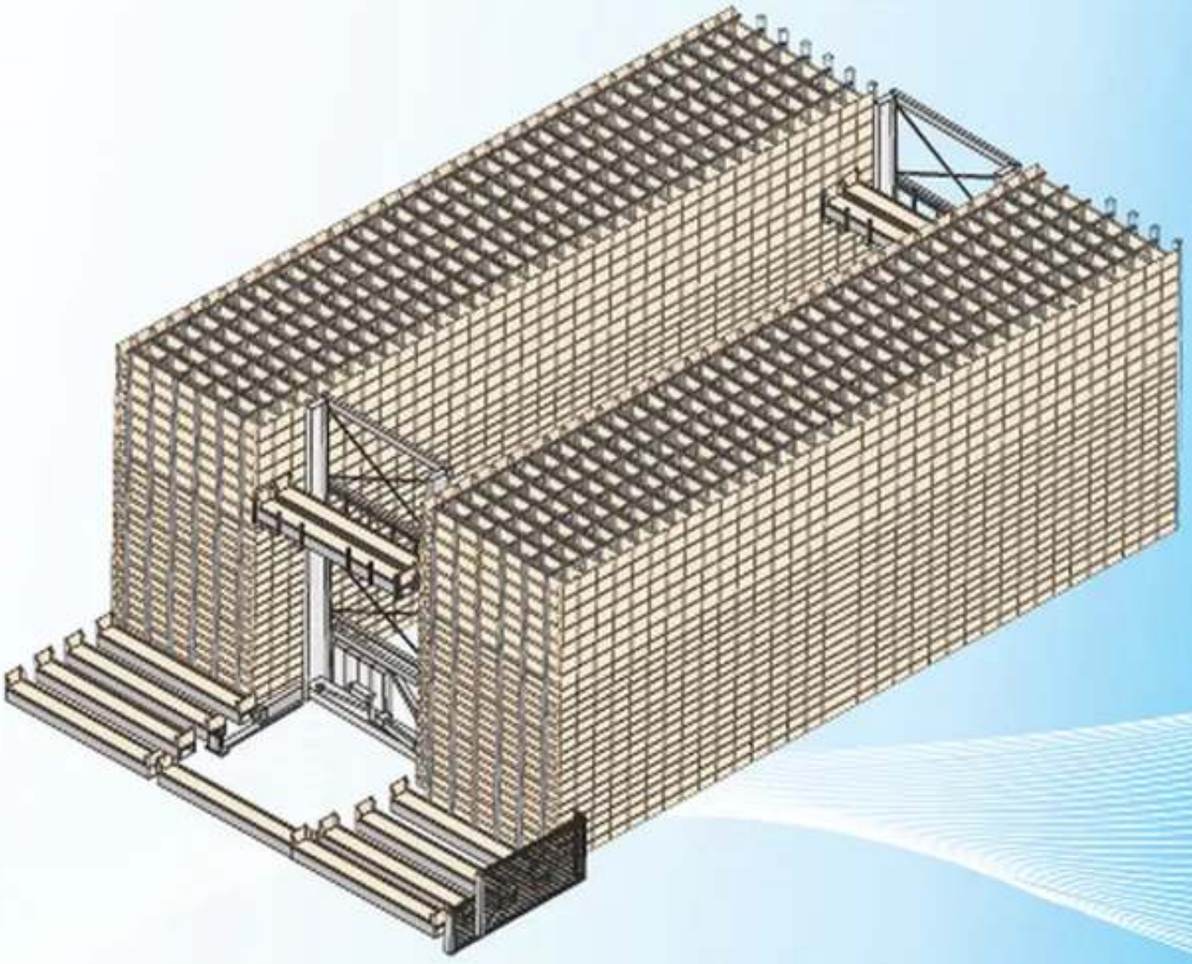
retrieval system for longer sections

Amatchless blend of swiftness & economy, Stolong is a perfect solution for storing and retrieving bars, building and facade profiles, metal fittings and sheet metal systems of varied sizes and quantities. It is most beneficial in steel and metal service centers and manufacturing, plastics and aluminium industries. A higher storage capacity, faster access and moderate investments are the main features of Stolong. Faster drives and a double pulling device of the gantry crane add to the efficiency of Stolong which guarantees economical material flow.

Perfect Automation

The ergonomically designed Stolong system offers perfect automation for the customer's storage requirement. It uses the random cassette principle. The storage systems can provide cycle times of varied speeds to suit his needs and bring the material to the operator. The gantry cranes in the system can achieve up to 50 double cycles per hour. Material can be retrieved from and carried to several workstations with more cycles than systems with permanent locations.

Stolong systems can reach height up to 25 m (82 ft). Stolong systems offer the following options – commissioning cranes, buffer tables with chain conveyors, bundling devices, truck routeing systems. Additional options of cassette dividers, inside linings and floor coverings are also available.



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OUR OTHER PRODUCT RANGE



Cantiliver Type Racking System



Heavy Duty Racking System



Motorised Heavy Duty Mobile System



Bin Type Fifo Racking System



Two Tier Racking System



Drawer Type Dia Storage System

Space Magnum

Total Storage Management (TSM)
Provider of Complete, custom-built Storage Solutions

SPACE MAGNUM
EQUIPMENTS PVT. LTD.

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